

Q1. HTTP Request–Response Cycle

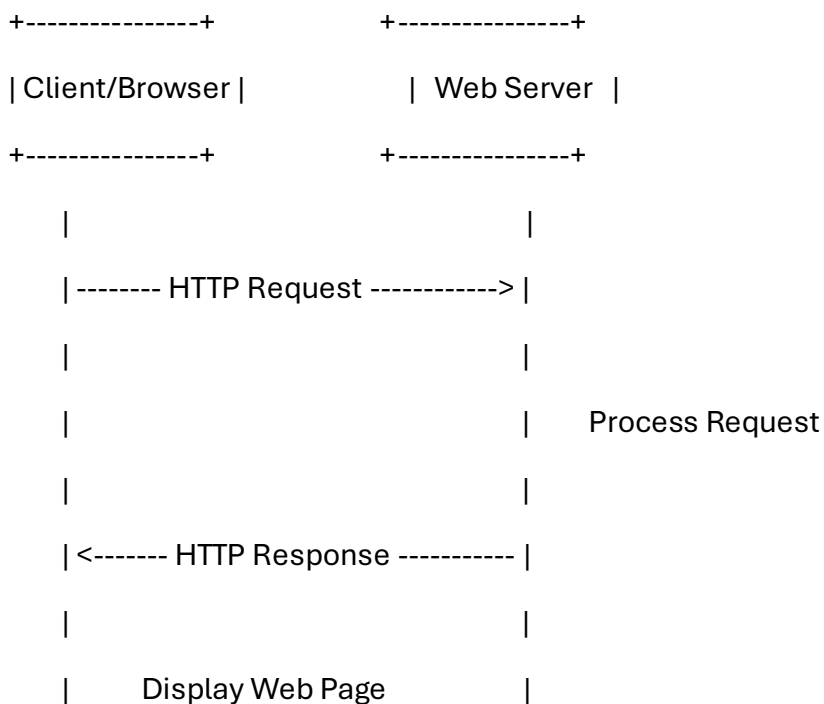
The **HTTP Request–Response Cycle** is the process through which a client, usually a web browser, communicates with a web server to obtain a webpage or other resource.

Ans :

The process works as follows:

1. The user enters a URL in the browser.
2. DNS finds the IP address associated with the domain name.
3. The browser sends an HTTP request to the web server.
4. The server receives and processes the request.
5. The server sends back an HTTP response with the required data.
6. The browser reads the response and displays the webpage.

The response may contain the requested HTML page, images, CSS files, or other resources.



Q2. Introduction to Web Technologies

Ans:

1. HTML

HTML (HyperText Markup Language) is used to build the basic structure of a webpage. It is used to add elements such as headings, paragraphs, links, images, tables, and forms.

2. CSS

CSS (Cascading Style Sheets) is used to style the content created using HTML. It controls the color, font, size, spacing, and layout of webpage elements.

3. JavaScript

JavaScript is used to make a webpage interactive. It can handle user actions, perform calculations, validate forms, and change webpage content dynamically.

4. Bootstrap

Bootstrap is a front-end framework that helps developers create responsive websites more easily. It provides a grid system and pre-designed components such as buttons, forms, cards, and navigation bars.

5. jQuery

jQuery is a JavaScript library that makes common JavaScript tasks easier, such as selecting HTML elements, handling events, creating animations, and working with AJAX.

6. React.js

React.js is a JavaScript library mainly used for developing user interfaces. It follows a component-based approach, which allows different parts of a webpage to be created and reused efficiently.

Q3. Client-Side vs. Server-Side Technologies

Ans :

Client-Side Technologies

Client-side technologies run in the user's web browser. They are mainly responsible for the appearance of the webpage and user interaction.

Examples: HTML, CSS, JavaScript, React.js.

Server-Side Technologies

Server-side technologies run on the web server. They process requests, perform business logic, work with databases, and send the required response to the client.

Examples: Java, Python, PHP, Node.js.

Client-Side	Server-Side
Runs in the user's browser	Runs on a web server
Handles webpage design and interaction	Handles processing and business logic
Mainly works with the user interface	Can communicate with databases
Examples: HTML, CSS, JavaScript	Examples: Java, Python, PHP

Q4. Block-Level Elements in HTML

Ans :

Block-level elements are HTML elements that generally begin on a new line and take up the available width of their parent element.

Main Characteristics:

- They usually start on a new line.
- They occupy the available horizontal space.
- They are commonly used to divide and organize webpage content.

Example - `<div>This is a division.</div>`

`<p>This is a paragraph.</p>`

`<h1>This is a heading.</h1>`

Other block-level elements include `<section>`, `<article>`, `<header>`, and `<footer>`.

Q5. Semantic HTML Elements

Ans :

Semantic elements are HTML elements that describe the type or purpose of the content they contain.

Some commonly used semantic elements are:

`<header>Header Content</header>`

`<nav>Navigation Links</nav>`

`<main>Main Content</main>`

`<section>Section Content</section>`

`<article>Article Content</article>`

`<footer>Footer Content</footer>`

Importance of Semantic Elements:

- They make HTML code easier to read and understand.
- They provide a better structure to the webpage.
- They improve accessibility.
- They help search engines understand the content more clearly.

- They make the code easier to maintain.

Unlike <div>, semantic elements provide information about the purpose of the content.

Q6. Creating a Table Using HTML

Ans :

```
<table border="1">

  <tr>

    <th>Sr. No.</th>

    <th>Name</th>

    <th>Designation</th>

    <th>Department</th>

  </tr>

  <tr>

    <td>1</td>

    <td>John</td>

    <td>Manager</td>

    <td>Sales</td>

  </tr>

  <tr>

    <td>2</td>

    <td>Smith</td>

    <td>Sr. Manager</td>

    <td>Marketing</td>

  </tr>

  <tr>

    <td>3</td>

    <td>Jane</td>

    <td>Manager</td>

    <td>HR</td>

  </tr>

</table>
```

Q7. Creating a Nested List Using HTML

Ans :

```
<ul>
  <li>Friends
    <ol>
      <li>Krishna</li>
      <li>Harsh</li>
      <li>Mohak</li>
      <li>Prachi</li>
    </ol>
  </li>
</ul>
```

Q8. Types of CSS

Ans :

There are mainly **three ways to apply CSS** to an HTML document.

1. Inline CSS

Inline CSS is written directly inside an HTML element using the style attribute.

Example - <p style="color: red;">Hello World</p>

2. Internal CSS

Internal CSS is written inside the <style> tag within the <head> section of an HTML document.

Example - <style> p { color: red; } </style>

3. External CSS

External CSS is written in a separate file with the .css extension and linked to the HTML document.

Example - <link rel="stylesheet" href="style.css">

External CSS is useful when the same style needs to be applied to more than one webpage

Q9. HTML, CSS and JavaScript

HTML, CSS, and JavaScript work together to create a complete webpage.

- **HTML** is responsible for creating the structure and content.
- **CSS** is responsible for the appearance and layout.

- **JavaScript** is responsible for interaction and dynamic behavior.

For example, HTML can create a button, CSS can change its color and size, and JavaScript can define what happens when the user clicks it.

Therefore, **HTML provides the structure, CSS provides the style, and JavaScript provides functionality and interaction.**

Q10. Short Answer

a. What is the role of Bootstrap in web development?

Bootstrap is a front-end framework that helps in developing responsive websites quickly. It provides a grid system and ready-to-use components such as buttons, forms, cards, and navigation bars.

b. What is jQuery and why was it widely used in web development?

jQuery is a JavaScript library that simplifies tasks such as DOM manipulation, event handling, animations, and AJAX. It became popular because it allowed developers to perform these tasks using less code.

c. What is React.js and what problem does it solve in frontend development?

React.js is a JavaScript library used to build user interfaces. It helps manage complex and dynamic webpages by dividing the interface into reusable components, making development and updates easier.